MENGYE CHEN, PH.D.

Norman, OK (open to relocation) · Portfolio: mcspace.work
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APPLIED AI SCIENTIST WHO DOESN'T TRUST AI

Machine Learning & LLM Systems · AI Agents & LLM Evaluation · Full-Stack AI Applications

Builds AI that forecasts floods, then makes it **earn trust**: benchmarks it against the simplest guess ("tomorrow looks like today") and co-authored a 2026 commentary on how AI can **look right for the wrong reasons**.

Ph.D. engineer with 10+ years building **machine-learning models, LLM-powered agents, and large-scale data pipelines**. Currently designs **LLM training and evaluation data** (coding tasks, reasoning traces, rubrics) and ships **full-stack AI applications** in Python, PyTorch, FastAPI, Node.js, and Hugging Face; 30+ peer-reviewed publications.

- LLM Evaluation, Rubrics & Reasoning Traces
- AI Agents & Tool-Using LLM Workflows
- Deep Learning (U-Net, LSTM, Source Separation)
- Full-Stack AI Apps & Interactive Dashboards
- Python Software Engineering & Automation
- Cloud, GPU & HPC Deployment
- Large-Scale Geospatial & Time-Series Pipelines
- Model Validation, Benchmarking & Uncertainty

TECHNICAL SKILLS

Languages: Python (PyTorch, scikit-learn, NumPy, Pandas, xarray, GeoPandas), JavaScript/Node.js, C++, SQL, R
LLM & AI: CrewAI multi-agent pipelines, tool calling, prompt design, vLLM, OpenAI & HF Inference APIs, ONNX
Engineering: FastAPI, Gradio, Leaflet, Docker, Git/GitHub Actions, AWS, Hugging Face Spaces (ZeroGPU), SLURM
Data: CREST/EF5, WRF, CMIP6, MRMS, GPM IMERG, GLDAS, NetCDF, Parquet/Arrow, GIS rasters, SQLite

PROFESSIONAL EXPERIENCE

ALIGNERR, Remote 07/2026 – Present

Applied AI Scientist (LLM Training & Evaluation)

Build expert training and evaluation data for LLM coding and engineering reasoning.

- Design complex programming-logic and algorithmic tasks that stress-test LLM reasoning, problem decomposition, and solution correctness.
- Author reasoning traces, reference solutions, and grading criteria that train LLMs in structured problem-solving, algorithm design, debugging, and engineering decisions.
- Audit LLM-generated code and reasoning for logic errors, edge cases, and failure modes.
- Reviewed ~100 AI-generated 2D-to-3D AutoCAD conversions, flagging missing labels and structural details.

UNIVERSITY OF OKLAHOMA, Norman, OK 01/2018 – Present

Researcher, Hydrometeorology & Remote Sensing Laboratory (HyDROS) **12/2023 – Present**

Lead AI-enabled flood-prediction research and build the software that delivers it to users.

- Co-developed **AQUAH**, an LLM- and vision-based agent that turns a plain-English flood request into a finished hydrologic study: it retrieves data, configures and runs the model, and writes the report (ICCV 2025 workshop).
- Built and deployed **CREST AI** (200+ commits), a FastAPI + Leaflet flood dashboard on a Docker Hugging Face Space: a CrewAI multi-agent pipeline (vLLM-served LLM, OpenAI fallback) parses a plain-language query, finds the basin and gauge, pulls MRMS forcing, runs CREST/EF5, and writes the report as hydrographs stream live.
- Developed **CREST_nowcast**, a DI-LSTM that nowcasts streamflow 1–6 h ahead from hourly MRMS radar and the latest USGS reading; benchmarked against a no-change baseline, trained on SLURM GPU nodes, served via API.
- Extended the C++ EF5 model engine with a native Apache Arrow/Parquet forcing reader, streaming output, per-pixel runoff calibration, and a lake module.
- Turned CONUS-wide CREST v3.0 calibration results into an interactive web viewer of ~9,000 USGS gauges with map-based selection and data download.
- Served as lead postdoc / Co-PI on an NSF INFEWS project spanning 5 universities and 1 national laboratory, translating research goals into executable technical tasks; led modeling for a DOE EPSCoR project.

Postdoctoral Researcher, Center for Analysis and Prediction of Storms (CAPS) **04/2021 – 06/2024**

- Built large-scale pipelines coupling CMIP6 projections, WRF dynamical downscaling, hydrologic models, and observations to quantify future flood, drought, and water-security risk.
- Engineered repeatable workflows for data-scarce regions (arid Peruvian Andes) and trained partner researchers in modeling and visualization (Tableau, Google Earth Studio).

Graduate Research Assistant, School of Civil Engineering & Environmental Science **01/2018 – 04/2021**

- Integrated U-Net deep-learning precipitation nowcasts and high-resolution forecasts into flood prediction for Hurricane Harvey, measuring gains in lead time and predictability (J. Hydrology 2022).
- Co-developed CREST-iMAP, a coupled hydrologic–hydraulic flood-mapping framework, and ran validation, sensitivity, and uncertainty tests across satellite, radar, gauge, and NWP inputs.

SELECTED AI & SOFTWARE PROJECTS

Independent projects, 2026. Live demos at mcspace.work*; code at* github.com/mchen15ouedu

- **AI Stem Splitter:** Gradio app on HF ZeroGPU that splits songs into 4/6/9 stems (Demucs, BS-RoFormer); an LLM assistant (Qwen2.5-7B with GPT-4o-mini and rule fallbacks) re-tunes settings from user quality complaints.
- **Rave Now:** Node.js show finder on two Docker HF Spaces: Twilio SMS/WhatsApp bot, SQLite, MusicBrainz checks, in-browser Whisper voice input, and a Qwen3-0.6B (ONNX) artist-name extractor.
- **Global Hydrological Zones:** Python pipeline (xarray, rasterio, GeoPandas, scikit-learn) condensing 20+ years of GPM IMERG/GLDAS data for 57,646 HydroBASINS watersheds into a nested PCA + k-means + Ward typology.

ADDITIONAL RELEVANT EXPERIENCE

WT CARE, LLC, **Founder & Chief Executive Officer** 01/2014 – 07/2017

Founded and led a ~10-person software and analytics company building route-optimization tools for weather and natural-hazard conditions; owned technical delivery, client requirements, budget, and timelines.

UNIVERSITY OF OKLAHOMA, **Instructor / Teaching Assistant (graduate courses)** 01/2019 – 05/2024

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN, **Research & Teaching Assistant** 01/2013 – 05/2014

Built vehicle-routing and scheduling optimization models; TA for FIN 418 Financial Modeling.

EDUCATION

Ph.D., Environmental Engineering, University of Oklahoma, 2021

M.S., Agricultural & Applied Economics, University of Illinois Urbana-Champaign, 2014

M.S., Environmental Engineering & Science, University of Illinois Urbana-Champaign, 2011

B.S., Environmental Systems Engineering, The Pennsylvania State University, 2010

SELECTED PUBLICATIONS

30+ peer-reviewed publications; full list on Google Scholar

- Yan, S., **Chen, M.**, Li, Z., et al. (2026). AI agent for hydrologic modeling: Definition, development, and application. Geophysical Research Letters, 53(13).
- Hong, Y., Gupta, H.V., Yong, B., **Chen, M.**, et al. (2026). HP Commentary: Neural equifinality and predictive mirages in AI-enabled hydrology and Earth system modelling. Hydrological Processes, 40, e70647.
- Yan, S., Li, Z., Zhu, S., Wen, Y., Zhang, M., **Chen, M.**, Cao, J., Hong, Y. (2025). AQUAH: Automatic Quantification and Unified Agent in Hydrology. ICCV 2025 Workshops (arXiv:2508.02936).
- **Chen, M.**, Li, Z., Gao, S., et al. (2022). A flood predictability study for Hurricane Harvey with the CREST-iMAP model using high-resolution quantitative precipitation forecasts and U-Net deep learning precipitation nowcasts. Journal of Hydrology, 612, 128168.
- **Chen, M.**, Li, Z., Vergara, H.J., et al. (2023). Conus-wide model calibration and validation for CRESTv3.0 – An improved Coupled Routing and Excess STorage distributed hydrological model. Journal of Hydrology, 626, 130333.

PROFESSIONAL SERVICE & LEADERSHIP

Associate Editor, Earth and Space Science · Lead Guest Editor, AGU special issue "Advances in Machine Learning for Earth Science" · Guest Editor, MDPI Atmosphere · Session Chair & Convener, AGU Fall Meeting